

# SimpleCRM Project Summary

**Live URL:** <https://power-agency-crm.web.app>

**Firebase Project:** power-agency-crm

**Last Updated:** 2026-02-23 (rev 2)

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## 1. Project Target

Build a **high-performance, lightweight CRM web application** for a power/marketing agency to manage:

- Contacts (people & companies) with rich profile data
- Sales pipeline management
- Task and calendar scheduling (with public booking pages)
- Email integration (Gmail API)
- AI-powered insights and briefings (Google Gemini)
- Team member management with role-based access
- Contact grouping / tagging
- Voice notes and live AI call assistant

### *Target User*

Agency staff (sales, account managers, admins) working from a browser no mobile app needed. The CRM should feel fast, modern, and require minimal training.

### *Delivery Plan*

Work was structured into 3 weeks:

| Week   | Focus                                            |
|--------|--------------------------------------------------|
| Week 1 | Core CRM: Contacts, Pipeline, Dashboard, Auth    |
| Week 2 | Email, Calendar, Todos, AI Features, Voice Notes |
| Week 3 | Polish, Bugs, Groups, Settings Integrations, UX  |

## 2. Tech Stack

| Layer              | Technology                                  |
|--------------------|---------------------------------------------|
| Frontend Framework | React 19 + TypeScript                       |
| Build Tool         | Vite 6                                      |
| Styling            | Tailwind CSS (CDN)                          |
| Backend / DB       | Firebase Firestore                          |
| Auth               | Firebase Auth (Email + Google OAuth)        |
| Hosting            | Firebase Hosting                            |
| AI                 | Google Gemini API (@google/genai)           |
| Email API          | Gmail API (via Google OAuth)                |
| Calendar API       | Google Calendar API (via Google OAuth)      |
| Icons              | lucide-react                                |
| Rich Text          | Custom RichTextEditor component             |
| Excel Import       | xlsx library                                |
| Voice              | Browser MediaRecorder API + Gemini Live API |

**No Redux, no React Router** all state lives in `App.tsx`, page switching via local state.

## 3. What Was Built

### *Week 1 Core CRM*

| Feature | Status | File |
|---------|--------|------|
|---------|--------|------|

|                                    |      |                   |
|------------------------------------|------|-------------------|
| Firebase Auth (Email + Google)     | Done | AuthPage.tsx      |
| Contact create / edit / delete     | Done | ContactForm.tsx   |
| Contact list with search & filters | Done | App.tsx           |
| Contact detail page (full profile) | Done | ContactDetail.tsx |
| People vs Company types            | Done | ContactForm.tsx   |
| Sales Pipeline (Kanban drag-drop)  | Done | Pipeline.tsx      |
| Dashboard with KPIs                | Done | Dashboard.tsx     |
| Sidebar navigation                 | Done | Sidebar.tsx       |

## ***Week 2 Email, Calendar, AI, Voice***

| Feature                           | Status | File                                        |
|-----------------------------------|--------|---------------------------------------------|
| Email inbox (Gmail API)           | Done   | EmailPage.tsx                               |
| Email compose with rich text      | Done   | EmailPage.tsx +<br>RichTextEditor.tsx       |
| Email labels / star / read        | Done   | EmailPage.tsx                               |
| Calendar event types              | Done   | CalendarPage.tsx                            |
| Public booking pages              | Done   | CalendarPage.tsx +<br>BookingPublicPage.tsx |
| Google Calendar sync              | Done   | CalendarPage.tsx                            |
| Todo / Task management            | Done   | TodoPage.tsx                                |
| AI daily briefing (Gemini)        | Done   | Dashboard.tsx                               |
| AI contact intelligence           | Done   | ContactDetail.tsx                           |
| Voice notes (record + transcribe) | Done   | ContactDetail.tsx                           |
| AI Live Call (real-time voice)    | Done   | App.tsx                                     |
| Contact import (Excel/XLSX)       | Done   | ImportModal.tsx                             |
| Custom fields in Settings         | Done   | SettingsPage.tsx                            |
| Team members + roles              | Done   | SettingsPage.tsx                            |

### Week 3 Polish & New Features

| Feature                          | Status | File                           |
|----------------------------------|--------|--------------------------------|
| Groups / Tags (full rewrite)     | Done   | GroupsPage.tsx                 |
| Group detail page                | Done   | GroupsPage.tsx                 |
| Bulk add/remove group members    | Done   | GroupsPage.tsx + App.tsx       |
| Email to group (compose prefill) | Done   | GroupsPage.tsx                 |
| Group analytics (status chart)   | Done   | GroupsPage.tsx                 |
| Contact field validations        | Done   | ContactForm.tsx                |
| Calendar Week view               | Done   | CalendarPage.tsx               |
| Calendar Day view                | Done   | CalendarPage.tsx               |
| Email Save Draft                 | Done   | App.tsx                        |
| Email Schedule Send              | Done   | EmailPage.tsx (dropdown arrow) |
| Settings integrations wired      | Done   | SettingsPage.tsx               |
| Inline errors (no more alerts)   | Done   | ContactDetail.tsx              |
| AI Live Call cancel bug fix      | Done   | App.tsx                        |
| Group Create modal bug fix       | Done   | App.tsx                        |

## 4. What Was Done (Session Work)

This is the detailed log of changes made in the development sessions.

### Bug Fixes

#### 1. AI Live Call Cancel Button Not Working

- **Problem:** `micStreamRef.getTracks()` was called directly instead of `micStreamRef.current.getTracks()`, causing a silent crash when stopping the call.
- **Fix:** Changed to `micStreamRef.current.getTracks()` in `stopLiveCall()` function in `App.tsx`.

#### 2. Group "Create" Button Did Nothing

- **Problem:** `isGroupModalOpen` state existed and was being set to `true` on click, but the modal JSX was never added to the component tree.

- **Fix:** Added the full group modal UI to `App.tsx` with name input, color picker, and Create/Save buttons.

### 3. `alert()` *Popups in ContactDetail*

- **Problem:** Two places used `alert()` for error messages (API key missing, file too large) which is poor UX.
- **Fix:** Added `chatError` and `attachError` state variables. Errors now display inline. `attachError` auto-dismisses after 4 seconds.

## *New Features Added*

### 4. *Contact Field Validations* (`ContactForm.tsx`)

Added validation in `handleSubmit`:

- **Phone:** regex pattern + minimum 7 digits
- **Website:** URL format check (optional field, only validated if filled)
- **Birthday:** must be yesterday or earlier (no future birthdays)

### 5. *Calendar Week & Day Views* (`CalendarPage.tsx`)

- Changed `meetingsViewMode` type from `'list' | 'calendar'` to `'list' | 'month' | 'week' | 'day'`
- **Week view:** 7-column SunSat grid, events shown as color pills per day
- **Day view:** 24-hour timeline with current hour highlighted, time-positioned events
- Both views support CRM + Google Calendar events and Prev/Today/Next navigation

### 6. *Email Save Draft + Gmail Persistence* (`App.tsx`, `firebase.ts`)

- Added `handleSaveDraft()` function async, writes draft to Firestore `email` collection
- Draft also added to local state immediately (optimistic update) so UI responds without waiting for Firestore
- Save button (floppy disk icon) added to compose toolbar
- **Gmail emails now cached in Firestore** after each fetch using Gmail message ID as Firestore doc ID (no duplicates on re-fetch)
- Firestore `email` listener loads all emails (Gmail-cached + drafts) on every page load no Gmail token needed after first connection
- Added `setDoc` to `firebase.ts` (real SDK + mock implementation) to support upsert by document ID
- Removed broken `sessionStorage` token approach Google OAuth tokens expire and cannot be reused across refreshes

### 7. *Groups Feature Full Rewrite* (`GroupsPage.tsx`)

Previous implementation was a simple list. Rewrote to include:

- **Group list** with color badges and member counts, clickable to open detail
- **Group Detail Page** (no route change, internal state) with 3 tabs:

- **Members tab:** search bar, checkbox selection, "Email Selected" / "Email All" buttons
- **Analytics tab:** summary cards, status breakdown bar chart, contact coverage percentage
- **Manage Members tab:** searchable list of all contacts with Add/Remove toggle per contact
- Group member updates dispatch `crm:group-toggle` custom event caught in `App.tsx` updates Firestore

## 8. Settings Integrations (`SettingsPage.tsx`)

- **Gmail:** real Connect button calls `onConnectGmail` prop shows green "Connected" badge when connected
- **Google Calendar:** real Connect button calls `onConnectGoogleCalendar` prop shows green "Connected" badge
- **7 others** (Zoom, Slack, LinkedIn, Google Meet, Office 365, Outlook, Teams): clicking shows amber "Coming Soon" toast for 3 seconds

## 5. What Is Pending

These items are known gaps but were **not explicitly requested** or **not yet prioritized**:

### Medium Priority

| Item                                    | Description                                                                                                                                                                          | Effort                        |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Email Trash/Spam/Archive folders</b> | Only Inbox and Sent show real Gmail API emails. Trash/Spam/Archive folders only show CRM-generated demo emails. Gmail API needs additional fetch calls for these folders.            | ~3 hours                      |
| <b>LinkedIn Profile data fetch</b>      | LinkedIn does not offer a public API for profile lookup without a formal LinkedIn Partner Program application. Currently no data is pulled this is a platform limitation, not a bug. | Requires LinkedIn partnership |

### Low Priority / Nice to Have

| Item                                         | Description                                                                            | Effort                   |
|----------------------------------------------|----------------------------------------------------------------------------------------|--------------------------|
| <b>Real Integrations</b> (Zoom, Slack, etc.) | 7 integrations show "Coming Soon". Each requires OAuth setup and API integration work. | 1-3 days per integration |

|                                  |                                                                                                            |          |
|----------------------------------|------------------------------------------------------------------------------------------------------------|----------|
| <b>File attachments in Email</b> | Compose window supports attaching files but Gmail API send with attachments needs MIME multipart encoding. | ~4 hours |
| <b>Contact photo upload</b>      | Currently accepts a URL string. Could add direct file upload to Firebase Storage.                          | ~3 hours |
| <b>Pipeline deal values</b>      | Pipeline shows contacts but no deal \$ values or revenue tracking.                                         | ~4 hours |
| <b>Mobile responsive polish</b>  | App works on mobile but some complex views (pipeline, calendar) are not optimized for small screens.       | ~1 day   |
| <b>Email pagination</b>          | Email inbox fetches a fixed number of emails. No load-more or pagination.                                  | ~2 hours |

## 6. Known Issues & Solutions

### *Issue: Gmail emails disappear on page refresh FIXED*

- **Root cause:** Google OAuth access tokens are short-lived and tied to the browser session. Saving them to sessionStorage does not help the token itself expires and any API call with it fails silently. There is no way to re-use an OAuth token across page refreshes without a full re-authentication flow.
- **Solution implemented:** Gmail emails are now **saved to Firestore** immediately after being fetched from the Gmail API. The Gmail message ID is used as the Firestore document ID (setDoc with merge: true) to prevent duplicates on re-fetch. The existing Firestore email listener loads all emails (Gmail-cached + drafts) on every page load no Gmail token needed after first connection.

### *Issue: Draft emails lost on refresh FIXED*

- **Root cause:** handleSaveDraft() only pushed to React local state (emails), which is cleared on refresh.
- **Solution implemented:** handleSaveDraft() now writes to Firestore email collection using addDoc. An optimistic local update is applied immediately so the UI responds without waiting for Firestore. The Firestore email listener then replaces the optimistic item with the real Firestore document.

### *Issue: LinkedIn data cannot be fetched automatically Platform Limitation*

- **Root cause:** LinkedIn's API requires a formal partnership and OAuth app approval. There is no public search endpoint.
- **Current behavior:** "Open LinkedIn" button deep-links to [https://linkedin.com/search/results/all/?keywords=\\${name}](https://linkedin.com/search/results/all/?keywords=${name}) so the user can manually copy data. This is by design.

### *Issue: Group member changes not persisting offline Not a Bug*

- **Root cause:** Group toggle uses Firestore updateDoc via App.tsx.
- **Status:** Firebase Firestore SDK handles offline caching automatically writes are queued locally and synced when connection restores. Works correctly already.

### *Issue: Gemini AI briefing fails silently if API key is missing*

- **Root cause:** GEMINI\_API\_KEY is injected at build time via Vite. If missing, the variable is undefined.
- **Solution:** Already handled chatError inline state in ContactDetail.tsx shows an error message in the UI. Dashboard briefing also shows a friendly error if the Gemini call fails.

## 7. File Structure

```
simplecrm/
  ■■■ index.html          # HTML entry point
  ■■■ index.tsx           # React entry + routing (AuthPage vs App)
  ■■■ App.tsx             # Main app – all state, all handlers (~3,000 lines)
  ■■■ firebase.ts        # Firebase init + mock backend fallback (exports: addDoc, setDoc, updateDoc)
  ■■■ constants.ts       # GROUP_COLORS, ACTIVITY_STYLES
  ■■■ utils.ts           # getFavicon, date helpers, audio utils
  ■■■ types.d.ts         # TypeScript declarations for Vite env
  ■■■ metadata.json      # App name, description, permissions
  ■■■ vite.config.ts     # Vite + React plugin, port 3000
  ■■■ tsconfig.json      # TypeScript config
  ■■■ package.json       # Dependencies + scripts
  ■■■ .env.local         # GEMINI_API_KEY, FIREBASE_CONFIG (not in git)
  ■■■ .firebaserc        # Firebase project: power-agency-crm
  ■■■ firebase.json      # Firebase hosting config (dist/, SPA rewrite)
  ■■■ components/
    ■ ■■■ AuthPage.tsx   # Login / Sign-up / Google sign-in
    ■ ■■■ Dashboard.tsx  # KPIs, AI briefing, activity feed
    ■ ■■■ Sidebar.tsx    # Navigation sidebar (collapsible)
    ■ ■■■ ContactForm.tsx # Create/edit contact form
    ■ ■■■ ContactDetail.tsx # Contact profile, activities, AI, voice
    ■ ■■■ Pipeline.tsx   # Kanban sales pipeline
    ■ ■■■ GroupsPage.tsx # Groups with detail, analytics, manage
    ■ ■■■ TodoPage.tsx   # Tasks with due dates
    ■ ■■■ CalendarPage.tsx # Events, booking pages, week/day/month views
    ■ ■■■ EmailPage.tsx  # Gmail inbox, compose, labels
    ■ ■■■ SettingsPage.tsx # Team, custom fields, integrations
    ■ ■■■ NoteItem.tsx   # Activity timeline item
    ■ ■■■ RichTextEditor.tsx # WYSIWYG editor
    ■ ■■■ Shared.tsx     # StatusBadge, DynamicInputList, WidgetItem
    ■ ■■■ RelatedCards.tsx # Related contacts/companies display
    ■ ■■■ BookingPublicPage.tsx # Public meeting booking form
    ■ ■■■ ImportModal.tsx # Excel contact import wizard
  ■■■ dist/              # Production build output (auto-generated)
```

## 8. Environment Setup

### *Required environment variables (.env.local):*

```
GEMINI_API_KEY=your_gemini_api_key_here
FIREBASE_CONFIG={"apiKey":"...", "authDomain":"...", "projectId":"power-agency-crm", "storageBucket":"...", "mes
```

### *Local development:*



```
cd simplecrm
npm install
npm run dev
# → http://localhost:3000
```

### ***Production build:***

```
npm run build
```

## **9. Deploy Command**

```
cd e:/Work/HelpAgency/simplecrm
npm run build && firebase deploy --only hosting
```

**Live URL:** <https://power-agency-crm.web.app>

## **Firestore Data Structure**

```
artifacts/
  power-agency-crm/
    users/
      {userId}/
        contacts/           # Contact and company records
        notes/              # Activity notes per contact (all types: Call, Email, Meeting, Task, Note)
        todos/              # Tasks with due dates
        tag_groups/         # Contact groups with colors
        event_types/        # Calendar event type templates
        scheduled_events/   # Booked meetings
        email/              # All emails: Gmail-cached (_fromGmail:true) + CRM drafts/sent
        custom_fields/      # User-defined custom fields (per contact type)
        pipelines/          # Sales pipeline definitions
        booking_pages/      # Public booking page configs
        team_members/       # Team member records with roles
```

**Note:** Collection names use snake\_case (e.g. tag\_groups, event\_types) match exactly when querying Firestore directly.